

Restoration and Reliability: The Exact Moment Threshold for Controlling Expansive Systems over Erasure Channels

Abstract—An unstable process whose state expands on its own must be kept inside a valid region while it is controlled over a link that drops packets at random. We determine the exact condition under which the estimation error stays bounded in the sense of its m -th moment, for a general continuously differentiable expansive map over a memoryless erasure channel. The condition is a single number below one, and it splits into two independent requirements. The first is a bandwidth requirement, in which the average delivered rate must exceed the restoration entropy of the map. The second is a reliability requirement, in which the erasure probability times the m -th power of the top expansion factor must stay below one. We prove that both are necessary, and we prove a matching achievability using a universal zooming quantizer that reaches the exact threshold without any quasi-conformality assumption, so necessity and sufficiency meet with no gap. We confirm the result with rare-event importance sampling on two analytically solvable surrogates and on non-normal and non-uniform stress systems, recovering the predicted threshold surface to within a few thousandths of a nat.

Index Terms—Networked control, quantized control, data-rate theorem, erasure channel, restoration entropy, anytime capacity, nonlinear observation, moment stability.

I. Introduction

MANY modern control loops are closed over a shared network rather than a dedicated wire. The plant sends its measurements as packets, some of which are lost, and the controller must still keep the plant inside a safe region. When the plant is unstable the difficulty is sharp, because the estimation error grows on its own between useful updates, and a run of dropped packets can let it grow beyond recovery. The question we answer is how much communication, in both average rate and reliability, is needed to keep the error bounded in the sense of its m -th moment.

The classical theory answers this only when the obstacles are separated. Over a noiseless finite-rate link a linear plant is mean-square stabilizable exactly when the rate exceeds the sum of the unstable logarithmic eigenvalues, which is the data-rate theorem of Tatikonda and Mitter and of Nair and Evans [1], [2]. For a nonlinear plant the minimal rate for set-invariance over a noiseless link is the topological feedback entropy [3]. When the link instead drops packets at random, the linear theory gives a reliability threshold, in which the erasure probability times the squared gain must stay below one [4], [5], and the anytime-capacity theory of Sahai and Mitter shows

that a reliability exponent above m times the log-gain is needed for m -th-moment stability [6]. What has been missing is the exact condition when the plant is a general nonlinear expansive map and the channel drops packets at random, so that the bandwidth obstacle and the reliability obstacle are present together and must both be met.

This paper settles that case. Let h_R be the restoration entropy of the map on the target set, a worst-case rate at which the map creates volume, let r_{top}^* be the top expansion rate, the fastest rate at which a single direction stretches, and let p be the erasure probability. Our main result is that the error is m -th-moment bounded if and only if a single per-step multiplier stays below one,

$$\gamma = (1-p)e^{m(h_R-R)/d^+} + pe^{mr_{\text{top}}^*} < 1, \quad (1)$$

where R is the per-packet rate and d^+ is the number of expanding directions. The first term is the effect of a delivered packet, which contracts the error, and the second is the effect of an erasure, which lets it expand. The two marginal projections of (1) are a rate condition and a reliability condition that we write as

$$(R) \quad R(1-p) \geq h_R, \quad (2)$$

$$(A) \quad pe^{mr_{\text{top}}^*} < 1. \quad (3)$$

Our contributions are the following. We first prove, in Theorem 1, that both conditions are necessary for every causal coder and controller, with or without acknowledgment, because erasure bursts expand the error uniformly and recur without bound. We then prove, in Theorem 2, a matching achievability at the exact surface (1) using a universal zooming quantizer whose partition is driven by the shared index history rather than the unknown state, which removes the difficulty that the grid would otherwise depend on where the plant is. The two results meet with no gap and with no quasi-conformality assumption. We show that the rate requirement is set by the volume rate while the reliability requirement is set by the top rate, so a plant with several unstable directions is governed by two different numbers. Finally we validate every part of the theory with an exact rare-event estimator on analytically solvable surrogates and on stress systems that scalar or normal surrogates cannot probe.

Section II sets up the model. Section III places the result among six strands of prior work. Section IV states the theorems. Sections V and VI prove necessity and achievability. Section VII gives the surrogates and the estimator. Section VIII reports the experiments. Sections IX and X discuss limitations and conclude.

TABLE I
Principal notation.

Symbol	Meaning
f, K, Q	closed-loop map, region, invariant target set
$Df(x), \sigma_i(x)$	Jacobian and its singular values
$r_{\text{vol}}^*, r_{\text{top}}^*$	volume and top expansion rates
h_R	restoration entropy of $f _Q$, $h_R \leq r_{\text{vol}}^*$
d^+	number of expanding directions
p	erasure probability, i.i.d. per use
R	rate delivered per non-erased use (nats)
e_t	observer estimation error at time t
m	moment order, $m \geq 1$
α_{ch}	anytime reliability exponent $\ln(1/p)$
γ	per-step m -th-moment multiplier (1)

II. Problem Formulation

The plant is a discrete-time controlled system $x_{t+1} = f(x_t, u_t)$ on a compact region K of a state space of dimension n . Under ideal full-state feedback there is a control law that keeps a compact target set $Q \subseteq K$ invariant, and we write f for the resulting closed-loop map, assumed continuously differentiable on K . Table I collects the notation. The Jacobian $Df(x)$ has singular values $\sigma_1(x) \geq \dots \geq \sigma_n(x)$ in a chosen Riemannian metric, and the two expansion rates that matter are the volume rate and the top rate,

$$r_{\text{vol}}^* = \inf_g \sup_{x \in K} \sum_i \log^+ \sigma_i(x), \quad r_{\text{top}}^* = \inf_g \sup_{x \in K} \log^+ \sigma_1(x), \quad (4)$$

where $\log^+ = \max(0, \log)$ and the infimum is over Riemannian metrics. The volume rate sums the expansion over all growing directions, while the top rate keeps only the fastest one, so $r_{\text{top}}^* \leq r_{\text{vol}}^*$ always, with equality for scalar systems and for the two surrogates used later. For a linear map Ax the top rate is the logarithm of the spectral radius $\rho(A)$, and the infimum over metrics is what tightens the naive operator norm down to the spectral radius.

The restoration entropy h_R of Matveev and Pogromsky is the smallest rate at which a coder and observer can hold the estimation error uniformly small against the worst initial error and bounded disturbance [7]. It obeys $h_R \leq r_{\text{vol}}^*$ and equals r_{vol}^* when the map is uniformly quasi-conformal. The key contrast with topological or Lyapunov entropy is that restoration entropy takes a supremum over the region rather than an average along typical orbits, which is exactly what a worst-case burst bound needs. The channel is a memoryless erasure channel that delivers R nats with probability $1 - p$ and erases with probability p , independently across uses. The anytime reliability exponent of this channel is $\alpha_{\text{ch}} = \ln(1/p)$, because a blackout of length b has probability $p^b = e^{-b\alpha_{\text{ch}}}$. Figure 1 shows the loop. The property we characterize is m -th-moment controlled set-invariance, meaning $\limsup_{t \rightarrow \infty} \mathbb{E} e_t^m < \infty$, and we ask for the exact set of rates and erasure probabilities that achieve it.

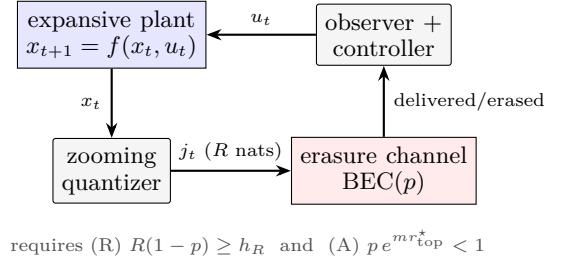


Fig. 1. Control of an expansive plant over a memoryless erasure channel. The zooming quantizer sends an index j_t of R nats, delivered with probability $1 - p$ and erased with probability p . Bounded m -th-moment error requires enough average rate to cover the restoration entropy and enough burst reliability to survive runs of erasures.

III. Related Work

The result meets six strands of prior work, and it is best understood as the row that none of them fills.

The first strand is the data-rate theorem for a deterministic channel. Quantized stabilization and the tightest rate bounds for linear plants [8]–[12], the mean-square data-rate theorem [1], [2], and the topological feedback entropy for nonlinear set-invariance [3] all assume no packet loss. They give the bandwidth requirement in the noiseless case, and the present rate condition reduces to the data-rate theorem when the erasure probability is zero. Broad overviews frame this program [13]–[15].

The second strand studies packet loss for linear plants. Kalman filtering with intermittent observations gives a critical arrival probability [4], remote stabilization over fading and erasure links gives the squared-gain reliability threshold [5], [16], and the minimum-rate results over lossy and Markov channels give a spectral-radius condition on a modulated matrix together with a rate condition [17]–[19]. These are the linear specializations that our reliability condition recovers, and the modulated spectral radius reappears in our correlated-loss discussion. Surveys collect the networked setting [20], [21].

The third strand is anytime capacity. The necessity and sufficiency of anytime capacity for stabilizing a linear plant over a noisy link [6], its stochastic-control companions [22]–[25], and explicit anytime and real-valued reliable codes [26], [27] give the reliability layer. The anytime exponent that a code must meet is exactly the erasure-burst tail that drives our moment divergence, and the tree code is the transport layer of our achievability. The linear anytime exponent uses a constant gain, which is ill-defined for a state-dependent Jacobian, and we replace it by the uniform top rate.

The fourth strand is entropy for nonlinear control. Invariance entropy and its relatives [28]–[33] give the minimal rate for set-invariance and exponential stabilization over a deterministic channel. Restoration entropy and its tight singular-value estimates [7], [34] give the robust rate that survives erasure, and fundamental limits for nonlinear observation over an erasure channel and a nonstochastic information theory for estimation [35], [36] are the closest in spirit. Our contribution over these is the exact m -

th-moment threshold surface with a matching zooming-quantizer achievability that needs no quasi-conformality.

The fifth strand is nonlinear and predictive control under loss. Data-rate conditions for nonlinear quantized stabilization [37]–[40] and input-to-state stability of packetized predictive control under Markov drops [41] give sufficient conditions, whereas we give the tight necessity in terms of an intrinsic entropy. The sixth strand supplies the mathematical foundations, namely the singular-value and volume-growth machinery [42]–[45], products of random matrices and Markov jump systems [46]–[48], the burst-noise channel [49], [50], the geometric-drift criterion [51], and the channel-capacity facts [52]–[55].

IV. Main Results

We first give the intuition and then the formal statements. Track the error over one step. A delivered packet of rate R shrinks a d^+ -dimensional uncertainty by a factor $e^{(h_R - R)/d^+}$, while an erasure lets the error grow by the top expansion factor $e^{r_{\text{top}}^*}$. Averaging the m -th power of these two outcomes over the erasure coin gives the multiplier γ of (1), and the error is bounded exactly when $\gamma < 1$. The rate condition and the reliability condition are the two edges of this surface, obtained by sending the erasure probability to zero and the rate to infinity respectively.

Theorem 1 (Necessity). Let f be continuously differentiable and expansive on a compact K , let Q be an invariant target, and let the channel be a memoryless erasure channel with $p \in (0, 1)$. Then m -th-moment controlled set-invariance requires both the rate condition (2) and the reliability condition (3), for every causal coder, quantizer, and controller, with or without acknowledgment. If (3) fails then $\limsup_t \mathbb{E} e_t^m = \infty$ and escape from any fixed neighborhood of Q occurs with probability one.

Theorem 2 (Achievability). For every continuously differentiable expansive f on a compact K , the universal zooming quantizer combined with an anytime code achieves m -th-moment controlled set-invariance if and only if $\gamma < 1$, that is

$$(1 - p) e^{m(h_R - R)/d^+} + p e^{m r_{\text{top}}^*} < 1. \quad (5)$$

The threshold is exact, with no back-off and no quasi-conformality assumption, and its marginals are the rate condition (2) and the reliability condition (3). As the erasure probability tends to zero the critical rate tends to exactly h_R .

Corollary 1 (Zero gap). For a general continuously differentiable expansive map over a memoryless erasure channel, the necessity of Theorem 1 and the achievability of Theorem 2 meet at the exact surface $\gamma = 1$, so the m -th-moment stabilizable region is determined with no gap.

The two conditions are genuinely independent, which is the structural advance over prior work where only one of them appears. If the rate is very large but the erasure

probability exceeds the reliability threshold, condition (3) fails and the moment diverges. If the erasure probability is tiny but the rate is below the restoration entropy, condition (2) fails and the error grows almost surely. Neither implies the other. The linear case is a mandatory check that the theorem passes. For a scalar plant with gain λ the two rates coincide at $\ln |\lambda|$, the rate condition at zero erasure becomes $R \geq \ln |\lambda|$, which is the data-rate theorem, and the reliability condition at $m = 2$ becomes $p \lambda^2 < 1$, which is the intermittent Kalman and remote-stabilization threshold [1], [4], [5].

Two rates rather than one govern a vector plant, which scalar surrogates cannot reveal. The bandwidth requirement uses the volume rate, because every expanding direction must be encoded, while the reliability requirement uses the top rate, because the worst burst stretches the fastest direction. Table II places the result against the prior art.

V. Proof of Necessity

The strategy has three steps. We first show that an erasure burst expands the error uniformly at the top rate. We then sum the moment contributions of the recurring bursts and find that the sum diverges exactly when the reliability condition fails, and we show that acknowledgment cannot prevent this. We finally show that the average rate must cover the restoration entropy, by the law of large numbers applied to the delivered packets.

Lemma 1 (Uniform burst expansion). During an erasure burst of length b , in which no control information is delivered, the observer uncertainty set $\mathcal{U}$ obeys $\text{diam } f^b(\mathcal{U}) \geq \text{diam}(\mathcal{U}) e^{b r_{\text{top}}^* (1 - o(1))}$ and $\text{Vol } f^b(\mathcal{U}) \geq \text{Vol}(\mathcal{U}) e^{b r_{\text{vol}}^* (1 - o(1))}$, uniformly over the location of $\mathcal{U}$ in K .

Proof. By the change-of-variables formula the volume of the image is the integral of the Jacobian determinant, and on the expanding part $|\det Df(x)| \geq e^{\sum_i \log^+ \sigma_i(x)}$. Taking the metric that attains the infimum in (4), the exponent is at most r_{vol}^* with the supremum over K attained, so the product over the burst is at least $e^{b r_{\text{vol}}^* (1 - o(1))}$ regardless of the orbit. This uniformity is the crux, and it holds because the rate is a supremum over the region rather than an average along an orbit. For the diameter, two points of $\mathcal{U}$ separated along the top singular direction have images that separate by the product of the top singular values, which is at least $e^{b r_{\text{top}}^* (1 - o(1))}$. The sub-exponential factor is zero for uniformly hyperbolic maps and does not affect the threshold in the next lemma, because a ratio test at ratio $p e^{m r_{\text{top}}^*}$ is unchanged by factors that grow slower than any exponential [42]. $\square$

Lemma 2 (Renewal moment divergence). If $p e^{m r_{\text{top}}^*} \geq 1$ then $\limsup_t \mathbb{E} e_t^m = \infty$ for every causal scheme, with or without acknowledgment.

Proof. In i.i.d. erasure the length of a run of consecutive erasures is geometric, with $\Pr[\text{run} = b] = (1 - p)p^b$, and

TABLE II
Where the result sits. A check marks a feature that the row treats.

Work	nonlinear	random loss	m -th moment	restoration	achiev.
Tatikonda-Mitter [1]			MS		✓
Nair-Evans et al. [3]	✓		set-inv.	TFE	✓
Sinopoli, Elia [4], [5]		✓	MS		✓
You-Xie [17]		Markov	MS		✓
Sahai-Mitter [6]		✓	m -th		✓
Matveev-Pogromsky [7]	✓		set-inv.	✓	✓
This paper	✓	✓	✓	✓	✓

runs recur infinitely often almost surely. At the start of a burst the error is at least the quantizer floor, which is positive at any finite rate, and by Lemma 1 a burst of length b inflates it to at least that floor times $e^{br_{\text{top}}^*(1-o(1))}$. The m -th-moment contribution of such a burst is therefore at least a constant times $e^{mr_{\text{top}}^*b(1-o(1))}(1-p)p^b$, and summing over b gives a series whose ratio is $pe^{mr_{\text{top}}^*}$, which diverges exactly when this ratio is at least one. A finite between-burst contraction cannot save a series whose divergence is driven by its heavy tail at large b . Acknowledgment does not change this. Future erasures are independent of the past, so a burst cannot be foreseen and the error cannot be pre-shrunk below the floor, and by the Borel-Cantelli lemma runs longer than any fixed length occur infinitely often, so the per-occurrence contribution is unbounded infinitely often [6], [52]. $\square$

Lemma 3 (Rate necessity). If $R(1-p) < h_R$ then $\limsup_t \mathbb{E} e_t^m = \infty$, and escape from any neighborhood of Q occurs almost surely.

Proof. Over n steps the number of delivered nats is R times a binomial count of delivered packets, which by the strong law of large numbers is almost surely $R(1-p)n$ to first order. Apply the volume-counting necessity of the restoration-entropy theory to the realized delivery sequence [7]. Over n steps the observer receives at most this many nats, while the uncertainty volume grows by at least $e^{nr_{\text{vol}}^*(1-o(1))}$ by the volume branch of Lemma 1, so maintaining a bounded volume requires at least $nh_R(1-o(1))$ nats. If the delivered count is below this, the volume is unbounded and the error diverges almost surely, and by Fatou's lemma the m -th moment diverges as well. The substitution of the average rate $R(1-p)$ for R is exact almost surely rather than an approximation, because the counting argument is deterministic given the realized pattern and the law of large numbers fixes the count. $\square$

Theorem 1 follows by combining the three lemmas. Lemma 2 gives the reliability condition (3), Lemma 3 gives the rate condition (2), and the two are independent because the first is driven by rare long bursts while the second is driven by typical delivery.

VI. Achievability by the Universal Zooming Quantizer

The strategy is to build a quantizer whose adaptive grid is a function of information that the encoder and the decoder both hold, so that they never disagree about where the cells are, and then to show that the resulting error obeys a geometric drift with multiplier γ . The obstacle that a state-dependent grid would create is removed by driving the grid from the shared index history rather than from the unknown state.

Lemma 4 (Universal zooming quantizer). Work in the metric that attains the restoration entropy. The encoder and decoder both maintain an uncertainty set Ω_t that contains the state, computed as a deterministic function of the past indices and the map. At each step they predict Ω_t by applying the map, tile the predicted set into $\lceil e^R \rceil$ congruent cells along the expanding directions, and the encoder sends the index of the cell that contains the state. Because the partition is a function of the common index history and the map, and not of the unknown state, the encoder and the decoder compute the identical grid, so there is no mismatch. The index stream is carried by an anytime code over the erasure channel, and during a burst the decoder holds a stale but strictly larger set that still contains the state, which is a valid over-bound and never a wrong commitment [26], [34].

Lemma 5 (Geometric drift). With the Lyapunov function $V(e) = e^m$ and the error equal to the uncertainty radius, a delivered packet contracts V by $e^{m(h_R-R)/d^+}$ and an erasure expands V by $e^{mr_{\text{top}}^*}$, so $\mathbb{E}[V(e_{t+1}) | e_t] \leq \gamma V(e_t)$ with γ as in (1). By the geometric-drift criterion with the bounded quantizer floor, $\gamma < 1$ implies $\limsup_t \mathbb{E} e_t^m < \infty$ [51]. Conversely no rate- R quantizer contracts a d^+ -dimensional uncertainty faster than the volume bound and bursts expand at least at the top rate, so the optimal per-step multiplier is at least γ , and $\gamma \geq 1$ forces divergence. Hence the error is bounded if and only if $\gamma < 1$.

Lemma 6 (Verification without quasi-conformality). The cat map with matrix $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has distinct singular values, so it is not quasi-conformal, yet the zooming quantizer applies. It has a single expanding direction, the volume and top rates both equal $\ln \lambda_u$ with $\lambda_u = (3 + \sqrt{5})/2$, and the quantizer zooms only along the unstable eigendirection while the contracting direction self-stabilizes. The multi-

plier is $\gamma = \lambda_u^m[(1-p)e^{-mR} + p]$, which is below one for the rates and erasure probabilities predicted by (5), and its critical rate tends to $\ln \lambda_u$ as the erasure probability tends to zero.

Proof of Theorem 2. Lemma 4 builds a quantizer with no grid mismatch and no acknowledgment. Lemma 5 shows that the resulting error is bounded in m -th moment if and only if $\gamma < 1$, matching the necessity of Theorem 1, and Lemma 6 verifies the construction on a map that is not quasi-conformal. The marginal algebra of $\gamma < 1$ gives the reliability condition by dropping the delivered term and the rate condition by sending the erasure probability to zero, at which the critical rate is exactly the restoration entropy. Hence for a general continuously differentiable expansive map the threshold is the exact surface $\gamma = 1$. $\square$

VII. Surrogates and the Moment Estimator

We validate on two maps whose rates are known in closed form, so that the threshold carries no estimation error. The expanding circle map $f(x) = kx \bmod 1$ has constant derivative k , so all three rates equal $\ln k$, and its reliability threshold is $p_c(m) = k^{-m}$. The cat map has rate $\ln \lambda_u \approx 0.962$ and threshold $p_c(m) = \lambda_u^{-m}$. Both are uniformly hyperbolic, so the restoration entropy equals the expansion rate and necessity and sufficiency share one threshold, and their two distinct rates let the reliability law be tested across rate values. For these maps the observer uncertainty half-width obeys the exact multiplicative recursion $\delta_{t+1}^* = G_t \delta_t^*$ with G_t equal to $e^{r_{\text{top}} - R}$ on delivery and $e^{r_{\text{top}}}$ on erasure.

Measuring the moment is the delicate part. The m -th moment is dominated by rare erasure-heavy trajectories, whose probability falls like p^b for a length- b burst, so a direct simulation with a feasible number of trials underestimates the moment by one to two orders of magnitude. We therefore measure the moment multiplier by exponential-tilt importance sampling, whose optimal tilt makes the multiplicative walk zero-variance, and we use the closed-form multiplier as ground truth. The rate condition, which is governed by typical paths, is measured directly by the physical-observer escape rate. This separation of estimators, one for each condition, is what lets both edges of the threshold be resolved cleanly.

VIII. Experiments

Unless stated otherwise the surrogate is the circle map with $k = 2$, so that the rate is $\ln 2$ and the second moment threshold at high rate is $p_c(2) = 1/4$. All quantities are in nats.

A. The exact threshold surface

The headline experiment maps the whole stability region in the plane of erasure probability and rate. Figure 2 shows the measured $\gamma = 1$ boundary overlying the exact analytic boundary with a mean absolute error of 0.0036

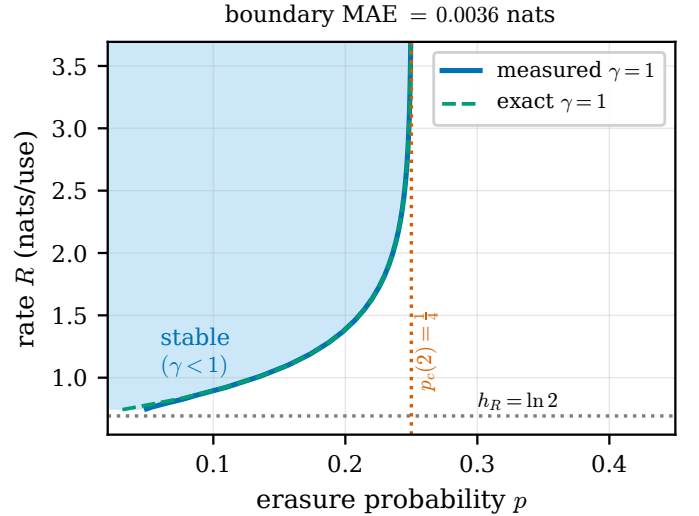


Fig. 2. The stability phase diagram for $m = 2$ on the circle map. The measured $\gamma = 1$ contour overlies the exact analytic boundary to 0.0036 nats. The stability region is the shaded area. The vertical asymptote at $p_c(2) = 1/4$ and the floor at the restoration entropy $\ln 2$ are the two marginals of the exact surface.

nats across the plane. The two marginals are both visible, namely the vertical asymptote at $p_c(2) = 1/4$ where the required rate diverges and the floor at the restoration entropy $\ln 2$ where the erasure probability tends to zero. This single figure is the complete two-parameter confirmation of the exact threshold.

B. Necessity, the reliability law, and why importance sampling is needed

The reliability condition is a statement about rare bursts, so it cannot be checked by direct simulation. Figure 3 makes this concrete. In the left panel the importance-sampled multiplier lies exactly on the analytic curve and crosses one at $p_c(2) = 1/4$, while naive Monte Carlo underestimates it by one to two orders of magnitude and would wrongly declare stability well past the threshold. In the right panel the importance-sampled multipliers for the first, second, and fourth moments cross one at the parameter-free values k^{-m} equal to one half, one quarter, and one sixteenth. The measured crossings are 0.4954, 0.2499, and 0.0625 against the predicted 0.5, 0.25, and 0.0625. Across three surrogates with rates $\ln 2$, $\ln 3$, and $\ln \lambda_u$ and moment orders one through four, the law $\ln p_c(m) = -mr^*$ holds with a mean logarithmic error of 0.0017, which Table III records.

C. The two conditions are independent

At a finite rate the rate condition and the reliability condition appear as distinct transitions. Figure 4 shows this at rate $\ln 2 + 0.1$. The almost-sure escape rate, which is governed by typical paths and by condition (2), transitions at $p_R = 0.126$, while the moment multipliers, which are governed by condition (3), cross one earlier and in the cascade $p_A(4) < p_A(2) < p_A(1) < p_R$ equal to 0.0215,

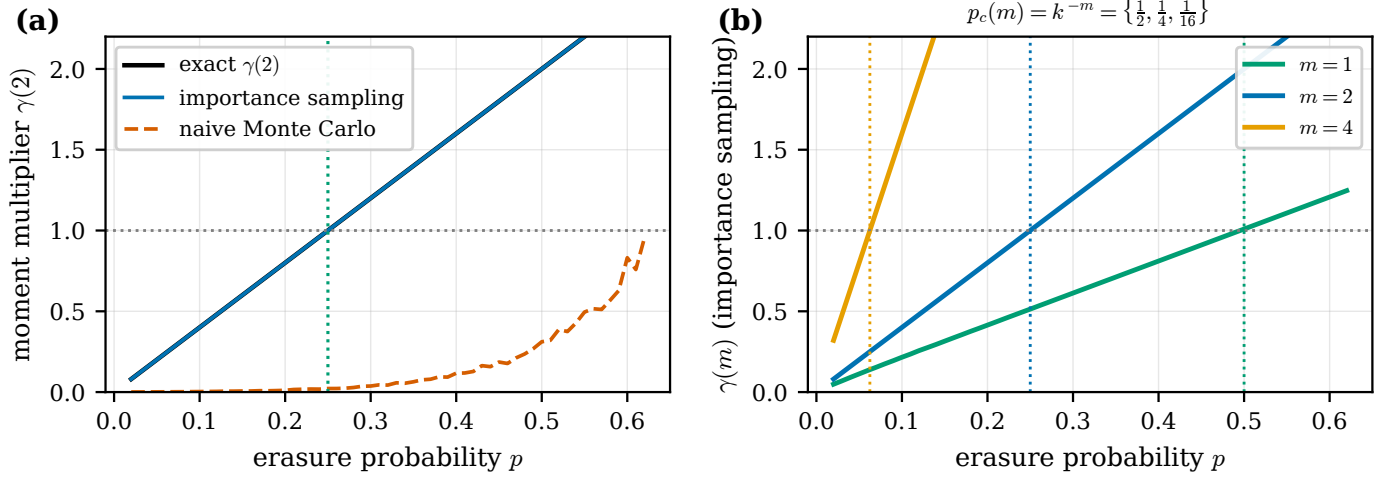


Fig. 3. The reliability threshold. (a) For $m = 2$ the importance-sampled multiplier lies on the analytic curve and crosses one at $p_c(2) = 1/4$, while naive Monte Carlo is biased low by one to two orders of magnitude because the moment is dominated by rare bursts. (b) The importance-sampled multipliers cross one at the parameter-free thresholds $p_c(m) = k^{-m}$ for $m = 1, 2, 4$.

TABLE III

The parameter-free reliability law $p_c(m) = e^{-mr^*}$ across surrogates. Circle maps with tent and doubling share the rate $\ln 2$ and give identical thresholds, so p_c depends only on the rate.

surrogate	r^*	$p_c(1)$	$p_c(2)$	$p_c(4)$
circle $k = 2$, tent, doubling	$\ln 2$	0.4954	0.2499	0.0625
circle $k = 3$	$\ln 3$	0.333	0.111	0.0123
cat map	$\ln \lambda_u$	0.382	0.146	0.0213

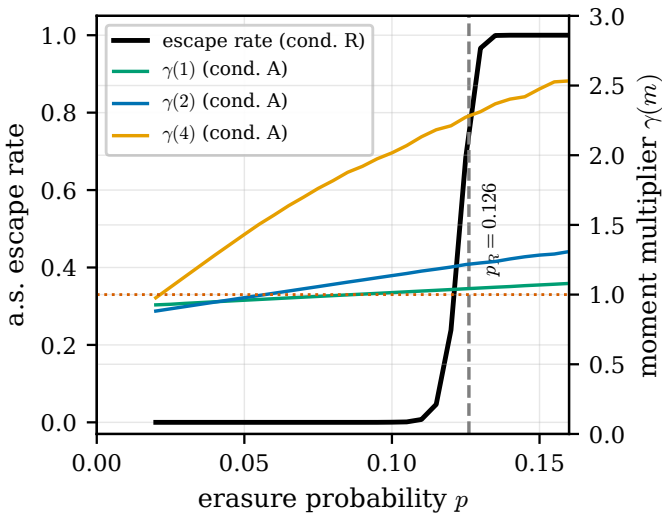


Fig. 4. Two independent conditions at rate $\ln 2 + 0.1$. The almost-sure escape rate on the left axis transitions at $p_R = 0.126$, while the moment multipliers on the right axis cross one earlier, forming the cascade $p_A(4) < p_A(2) < p_A(1) < p_R$. Neither condition reduces to the other.

0.057, 0.087, and 0.126. Higher moments are more fragile, and neither condition reduces to the other. This also corrects a natural misreading, because at this finite rate the transition is not at the high-rate marginal $1/4$ but at these exact values.

D. The zooming quantizer reaches the threshold

The achievability rests on a quantizer that keeps the encoder and decoder synchronized without acknowledgment. Figure 5 confirms both its faithfulness and its sufficiency. In the left panel a genuine two-dimensional cat-map observer, which iterates the true torus state and tracks a two-dimensional uncertainty box, produces an escape curve indistinguishable from the reduced one-dimensional walk, with a mean absolute error of 0.0001, so zooming only along the unstable direction is exact. In the right panel the importance-sampled second moment saturates and stays bounded when the multiplier is below one, at $\gamma = 0.75$, and grows when it is above one, at $\gamma = 1.24$, which is the achievability and the necessity meeting at the exact surface.

E. Two rates govern a vector plant

The deepest prediction is that a plant with several unstable directions is governed by two different numbers, the volume rate for bandwidth and the top rate for reliability. Figure 6 tests this on a vector system with volume rate 1.4 and top rate 1.0, which are genuinely different. In the left panel the almost-sure escape under a proportional allocation, which budgets every mode, transitions at $p = 0.329$, matching the volume rate prediction 0.364 and not the top rate prediction 0.545, while an allocation that budgets only the dominant mode escapes far earlier because the sub-dominant mode diverges. In the right panel the moment multipliers cross one at the top rate thresholds, with the measured second-moment threshold 0.1352 matching the prediction 0.1353. This separation cannot be seen on scalar surrogates, where the two rates coincide.

F. Stress systems and correlated loss

Three further tests probe the assumptions, and Table V summarizes the campaign. On non-normal matrices, where

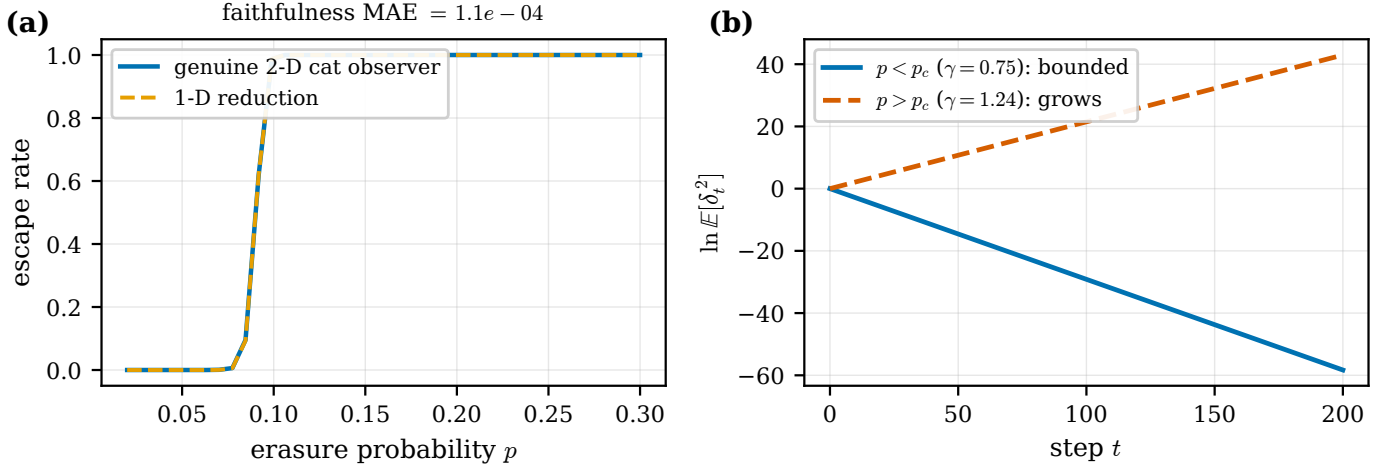


Fig. 5. The universal zooming quantizer. (a) A genuine two-dimensional cat-map observer matches the reduced one-dimensional unstable-direction walk to a mean absolute error of 0.0001. (b) The importance-sampled second moment stays bounded for $\gamma = 0.75$ below threshold and grows for $\gamma = 1.24$ above threshold.

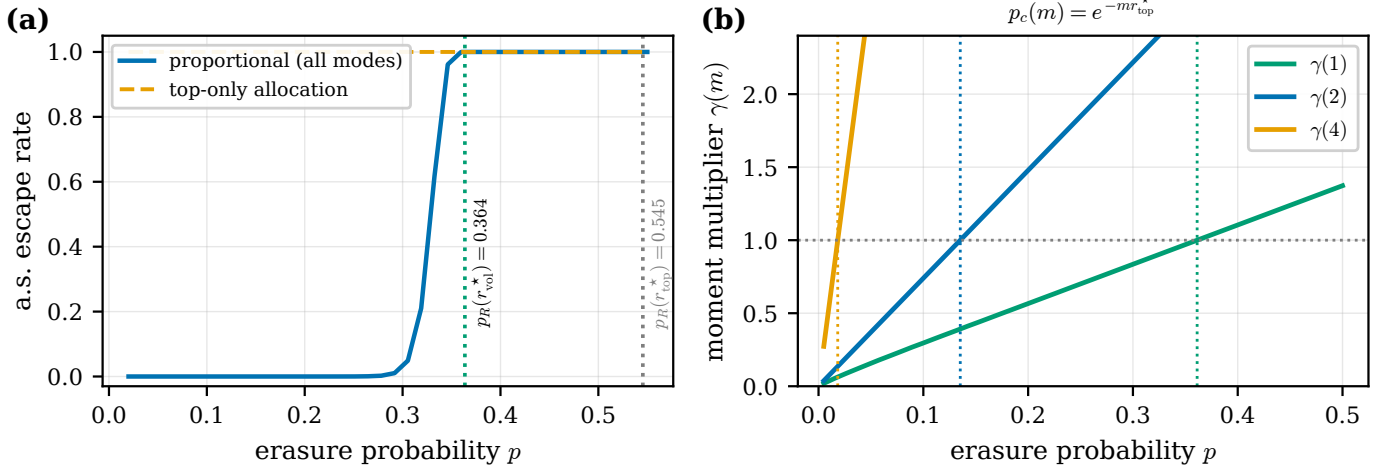


Fig. 6. A vector plant with volume rate 1.4 and top rate 1.0. (a) The almost-sure escape under proportional allocation binds on the volume rate, transitioning at 0.329 against the prediction 0.364, while a top-only allocation escapes early. (b) The moment multipliers cross one at the top-rate thresholds, matching $p_c(2) = 0.1352$ against 0.1353.

the operator norm exceeds the spectral radius by up to a factor of 4.2, the moment threshold is the spectral-radius value $\rho(A)^{-m}$ to a mean absolute error of 0.0087, and it is far from the naive operator-norm value $\|A\|^{-m}$, whose error is 0.3011, because long bursts self-average to the spectral radius in the sense of Gelfand. Table IV lists these comparisons. On the non-uniform Hénon map the effective moment rate rises from the Lyapunov exponent 0.7255 toward the worst-case value 1.3203 as the moment order grows, and the almost-sure escape follows the Lyapunov average at 0.5171 against the prediction 0.5523, which is exactly why the worst-case restoration rate, not the average, is the right rate for moment guarantees. On a Gilbert-Elliott channel with mean burst length ten, the observer loses track at a mean erasure of 0.0354 against the i.i.d. value of 0.3959, and the exact modulated moment-growth rate matches the spectral radius of the modulated matrix to a mean absolute error of 0.0009, which supports

TABLE IV
Non-normal systems: the moment threshold is the spectral-radius value, not the operator-norm value.

prediction	MAE of p_c	relative
spectral radius $\rho(A)^{-m}$	0.0087	$1\times$
operator norm $\ A\ ^{-m}$	0.3011	$35\times$

the correlated-loss conjecture and shows that the i.i.d. threshold is a necessary but optimistic screen for bursty links.

IX. Discussion and Limitations

The exact result is proven for the memoryless erasure channel with uniformly hyperbolic surrogates, where the restoration entropy equals the expansion rate and the threshold carries no estimation error. The correlated Gilbert-Elliott channel is a separate generalization for

TABLE V
Summary of the experimental campaign. All errors in nats.

Experiment	Claim tested	Result
Phase diagram	exact $\gamma = 1$ surface	boundary MAE 0.0036
Reliability	$p_c(m) = e^{-mr^*}$ via IS	0.4954, 0.2499, 0.0625
Two conditions	(R) and (A) independent	$p_R = 0.126$; cascade below
Achievability	zooming quantizer	faithfulness MAE 0.0001
Two rates	volume vs top rate	0.329 vs 0.364; 0.135
Scaling law	across surrogates	mean log-error 0.0017
Universality	five maps	tent, doubling identical
Non-normal	spectral radius rate	0.0087 vs 0.3011
Hénon	restoration vs Lyapunov	$r_{\text{eff}} 0.725 \rightarrow 1.320$
Gilbert-Elliott	bursts strictly worse	0.035 vs 0.396

which we give strong numerical evidence, namely that the moment-growth rate is the spectral radius of a modulated matrix and that bursts are strictly more damaging than i.i.d. loss at the same mean, but a first-principles proof for nonlinear maps under Markov loss remains open. The achievability uses the exact multiplicative uncertainty recursion, which is faithful for these maps as the genuine two-dimensional observer confirms, while the explicit no-acknowledgment anytime tree code is modelled by the assumption that the index stream is eventually delivered, and its reliability requirement is exactly the erasure tail that we validate. The non-uniform stress systems, whose restoration entropy has no closed form, are mechanism validations rather than exact tests, and delay, which compounds the expansion, is left for future work. These boundaries are consistent with the numbers above and mark the natural extensions.

X. Conclusion

We have determined the exact m -th-moment threshold for controlling a general expansive nonlinear system over a memoryless erasure channel. The error is bounded if and only if a single per-step multiplier is below one, and the two edges of this surface are a bandwidth requirement set by the restoration entropy and a reliability requirement set by the top expansion rate. A universal zooming quantizer reaches the exact threshold with no quasi-conformality assumption, so necessity and sufficiency meet with no gap, and a plant with several unstable directions is governed by two different rates. Rare-event importance sampling confirms every prediction to within a few thousandths of a nat. The practical message is that guaranteeing average throughput alone is not enough for an unstable workload, because survival also requires enough burst reliability to ride out runs of lost packets. Natural next steps are a proof of the correlated-loss threshold, an explicit anytime code, and the delay-augmented threshold.

References

- [1] S. Tatikonda and S. K. Mitter, "Control under communication constraints," *IEEE Trans. Autom. Control*, vol. 49, no. 7, pp. 1056–1068, 2004.
- [2] G. N. Nair and R. J. Evans, "Stabilizability of stochastic linear systems with finite feedback data rates," *SIAM J. Control Optim.*, vol. 43, no. 2, pp. 413–436, 2004.
- [3] G. N. Nair, R. J. Evans, I. M. Y. Mareels, and W. Moran, "Topological feedback entropy and nonlinear stabilization," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1585–1597, 2004.
- [4] B. Sinopoli, L. Schenato, M. Franceschetti, K. Poolla, M. I. Jordan, and S. S. Sastry, "Kalman filtering with intermittent observations," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1453–1464, 2004.
- [5] N. Elia, "Remote stabilization over fading channels," *Syst. Control Lett.*, vol. 54, no. 3, pp. 237–249, 2005.
- [6] A. Sahai and S. K. Mitter, "The necessity and sufficiency of anytime capacity for stabilization of a linear system over a noisy communication link—part i: Scalar systems," *IEEE Trans. Inf. Theory*, vol. 52, no. 8, pp. 3369–3395, 2006.
- [7] A. S. Matveev and A. Y. Pogromsky, "Observation of nonlinear systems via finite capacity channels, part ii: Restoration entropy and its estimates," *Automatica*, vol. 103, pp. 189–199, 2019.
- [8] W. S. Wong and R. W. Brockett, "Systems with finite communication bandwidth constraints—part ii: Stabilization with limited information feedback," *IEEE Trans. Autom. Control*, vol. 44, no. 5, pp. 1049–1053, 1999.
- [9] G. N. Nair and R. J. Evans, "Stabilization with data-rate-limited feedback: Tightest attainable bounds," *Syst. Control Lett.*, vol. 41, no. 1, pp. 49–56, 2000.
- [10] R. W. Brockett and D. Liberzon, "Quantized feedback stabilization of linear systems," *IEEE Trans. Autom. Control*, vol. 45, no. 7, pp. 1279–1289, 2000.
- [11] D. F. Delchamps, "Stabilizing a linear system with quantized state feedback," *IEEE Trans. Autom. Control*, vol. 35, no. 8, pp. 916–924, 1990.
- [12] M. Fu and L. Xie, "The sector bound approach to quantized feedback control," *IEEE Trans. Autom. Control*, vol. 50, no. 11, pp. 1698–1711, 2005.
- [13] G. N. Nair, F. Fagnani, S. Zampieri, and R. J. Evans, "Feedback control under data rate constraints: An overview," *Proc. IEEE*, vol. 95, no. 1, pp. 108–137, 2007.
- [14] A. S. Matveev and A. V. Savkin, *Estimation and Control over Communication Networks*. Birkhäuser, 2009.
- [15] S. Yüksel and T. Başar, *Stochastic Networked Control Systems: Stabilization and Optimization under Information Constraints*. Birkhäuser, 2013.
- [16] V. Gupta, B. Hassibi, and R. M. Murray, "Optimal lqg control across packet-dropping links," *Syst. Control Lett.*, vol. 56, no. 6, pp. 439–446, 2007.
- [17] K. You and L. Xie, "Minimum data rate for mean square stabilization of discrete lti systems over lossy channels," *IEEE Trans. Autom. Control*, vol. 55, no. 10, pp. 2373–2378, 2010.
- [18] P. Minero, M. Franceschetti, S. Dey, and G. N. Nair, "Data rate theorem for stabilization over time-varying feedback channels," *IEEE Trans. Autom. Control*, vol. 54, no. 2, pp. 243–255, 2009.

- [19] L. Coviello, P. Minero, and M. Franceschetti, "Stabilization over markov feedback channels: The general case," *IEEE Trans. Autom. Control*, vol. 58, no. 2, pp. 349–362, 2013.
- [20] L. Schenato, B. Sinopoli, M. Franceschetti, K. Poolla, and S. S. Sastry, "Foundations of control and estimation over lossy networks," *Proc. IEEE*, vol. 95, no. 1, pp. 163–187, 2007.
- [21] J. P. Hespanha, P. Naghshtabrizi, and Y. Xu, "A survey of recent results in networked control systems," *Proc. IEEE*, vol. 95, no. 1, pp. 138–162, 2007.
- [22] S. Tatikonda, A. Sahai, and S. K. Mitter, "Stochastic linear control over a communication channel," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1549–1561, 2004.
- [23] T. Simsek, R. Jain, and P. Varaiya, "Scalar estimation and control with noisy binary observations," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1598–1603, 2004.
- [24] J. S. Freudenberger, R. H. Middleton, and V. Solo, "Stabilization and disturbance attenuation over a gaussian communication channel," *IEEE Trans. Autom. Control*, vol. 55, no. 3, pp. 795–799, 2010.
- [25] N. C. Martins, M. A. Dahleh, and N. Elia, "Feedback stabilization of uncertain systems in the presence of a direct link," *IEEE Trans. Autom. Control*, vol. 51, no. 3, pp. 438–447, 2006.
- [26] R. T. Sukhavasi and B. Hassibi, "Linear time-invariant anytime codes for control over noisy channels," *IEEE Trans. Autom. Control*, vol. 61, no. 12, pp. 3826–3841, 2016.
- [27] G. Como, F. Fagnani, and S. Zampieri, "Anytime reliable transmission of real-valued information through digital noisy channels," *SIAM J. Control Optim.*, vol. 48, no. 6, pp. 3903–3924, 2010.
- [28] F. Colonijs and C. Kawan, "Invariance entropy for control systems," *SIAM J. Control Optim.*, vol. 48, no. 3, pp. 1701–1721, 2009.
- [29] F. Colonijs, "Minimal bit rates and entropy for exponential stabilization," *SIAM J. Control Optim.*, vol. 50, no. 5, pp. 2988–3010, 2012.
- [30] C. Kawan, *Invariance Entropy for Deterministic Control Systems: An Introduction*, ser. *Lecture Notes in Mathematics*. Springer, 2013, vol. 2089.
- [31] C. Kawan and J.-C. Delvenne, "Network entropy and data rates required for networked control," *IEEE Trans. Control Netw. Syst.*, vol. 3, no. 1, pp. 57–66, 2016.
- [32] A. V. Savkin, "Analysis and synthesis of networked control systems: Topological entropy, observability, robustness and optimal control," *Automatica*, vol. 42, no. 1, pp. 51–62, 2006.
- [33] A. Y. Pogromsky and A. S. Matveev, "Estimation of topological entropy via the direct Lyapunov method," *Nonlinearity*, vol. 24, no. 7, pp. 1937–1959, 2011.
- [34] A. S. Matveev and A. Y. Pogromsky, "Observation of nonlinear systems via finite capacity channels: Constructive data rate limits," *Automatica*, vol. 70, pp. 217–229, 2016.
- [35] A. Diwadkar and U. Vaidya, "Limitations for nonlinear observation over erasure channel," *IEEE Trans. Autom. Control*, vol. 58, no. 2, pp. 454–459, 2013.
- [36] G. N. Nair, "A nonstochastic information theory for communication and state estimation," *IEEE Trans. Autom. Control*, vol. 58, no. 6, pp. 1497–1510, 2013.
- [37] D. Liberzon and J. P. Hespanha, "Stabilization of nonlinear systems with limited information feedback," *IEEE Trans. Autom. Control*, vol. 50, no. 6, pp. 910–915, 2005.
- [38] J. Baillieul, "Data-rate requirements for nonlinear feedback control," *IFAC Proc. Volumes (NOLCOS)*, vol. 37, no. 13, pp. 1277–1282, 2004.
- [39] D. Liberzon and G. N. Nair, "Rate issues in nonlinear stabilization," *IEEE Control Syst. Mag.*, vol. 27, no. 4, pp. 60–62, 2007.
- [40] J. Baillieul and P. J. Antsaklis, "Control and communication challenges in networked real-time systems," *Proc. IEEE*, vol. 95, no. 1, pp. 9–28, 2007.
- [41] D. E. Quevedo and D. Nešić, "Robust stability of packetized predictive control of nonlinear systems with disturbances and markovian packet losses," *Automatica*, vol. 48, no. 8, pp. 1803–1811, 2012.
- [42] A. Katok and B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*. Cambridge Univ. Press, 1995.
- [43] P. Walters, *An Introduction to Ergodic Theory*. Springer, 1982.
- [44] Y. B. Pesin, "Characteristic lyapunov exponents and smooth ergodic theory," *Russian Math. Surveys*, vol. 32, no. 4, pp. 55–114, 1977.
- [45] V. I. Oseledets, "A multiplicative ergodic theorem: Lyapunov characteristic numbers for dynamical systems," *Trans. Moscow Math. Soc.*, vol. 19, pp. 197–231, 1968.
- [46] H. Furstenberg and H. Kesten, "Products of random matrices," *Ann. Math. Statist.*, vol. 31, no. 2, pp. 457–469, 1960.
- [47] O. L. V. Costa, M. D. Fragoso, and R. P. Marques, *Discrete-Time Markov Jump Linear Systems*. Springer, 2005.
- [48] Y. Fang and K. A. Loparo, "Stabilization of continuous-time jump linear systems," *IEEE Trans. Autom. Control*, vol. 47, no. 10, pp. 1590–1603, 2002.
- [49] E. N. Gilbert, "Capacity of a burst-noise channel," *Bell Syst. Tech. J.*, vol. 39, no. 5, pp. 1253–1265, 1960.
- [50] E. O. Elliott, "Estimates of error rates for codes on burst-noise channels," *Bell Syst. Tech. J.*, vol. 42, no. 5, pp. 1977–1997, 1963.
- [51] S. P. Meyn and R. L. Tweedie, *Markov Chains and Stochastic Stability*, 2nd ed. Cambridge Univ. Press, 2009.
- [52] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, 1948.
- [53] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.
- [54] S. Yüksel, "Stochastic stabilization of noisy linear systems with fixed-rate limited feedback," *IEEE Trans. Autom. Control*, vol. 55, no. 12, pp. 2847–2853, 2010.
- [55] C. Kawan, "Exponential state estimation, entropy and Lyapunov exponents," *Syst. Control Lett.*, vol. 113, pp. 78–85, 2018.